

Limit Laws and the Squeeze Theorem

MATH 1334

September 30, 2026

Recap: limits

$$\lim_{x \rightarrow a} f(x), \quad \lim_{x \rightarrow a^-} f(x), \quad \lim_{x \rightarrow a^+} f(x).$$

Table method Graph method Case method

Example. For $f(x) = \frac{x^2 - 4}{|x - 2|}$, use cases:

$$f(x) = \begin{cases} -(x + 2), & x < 2, \\ x + 2, & x > 2. \end{cases}$$

Thus $\lim_{x \rightarrow 2^-} f(x) = -4$ and $\lim_{x \rightarrow 2^+} f(x) = 4$, so the two-sided limit DNE.

Issues. These approaches can be slow; tables and sketches may be imprecise and do not by themselves prove a general limit.

Question. How do we systematically compute limits?

Section 2.3: limit laws

Theorem (Limit laws). Suppose $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, where L, M are real numbers. Let c be a real constant and n a positive integer.

Constant:	$\lim_{x \rightarrow a} c = c.$
Identity:	$\lim_{x \rightarrow a} x = a.$
Sum:	$\lim_{x \rightarrow a} (f(x) + g(x)) = L + M.$
Difference:	$\lim_{x \rightarrow a} (f(x) - g(x)) = L - M.$
Scalar multiple:	$\lim_{x \rightarrow a} cf(x) = cL.$
Product:	$\lim_{x \rightarrow a} f(x)g(x) = LM.$
Quotient:	$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0.$
Power:	$\lim_{x \rightarrow a} (f(x))^n = L^n.$
Root:	$\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{L}.$

For even roots, require $L \geq 0$ and $f(x) \geq 0$ for the nearby inputs. Odd roots allow negative values. The same laws apply to one-sided limits.

Using the limit laws

Example 1. Find $\lim_{x \rightarrow 2} (3x^2 - 2x + 1)$.

Limit laws.

$$\begin{aligned}\lim_{x \rightarrow 2} (3x^2 - 2x + 1) &= 3 \left(\lim_{x \rightarrow 2} x \right)^2 - 2 \lim_{x \rightarrow 2} x + 1 \\ &= 3(2)^2 - 2(2) + 1 = 9.\end{aligned}$$

This uses the sum, difference, scalar multiple, power, constant, and identity laws.

Example 2. Suppose

$$\lim_{x \rightarrow a} f(x) = 2 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 3.$$

Find

$$\lim_{x \rightarrow a} \frac{\sqrt{f(x) - 1}}{(g(x))^2}.$$

Root and quotient laws. The radicand tends to $2 - 1 = 1 > 0$, and the denominator tends to $3^2 = 9 \neq 0$. Therefore

$$\lim_{x \rightarrow a} \frac{\sqrt{f(x) - 1}}{(g(x))^2} = \frac{\sqrt{2 - 1}}{3^2} = \frac{1}{9}.$$

Polynomials and rational functions

Corollary. Let f be a polynomial or rational function. If a is in the domain of f , then

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Proof. For a polynomial $p(x) = c_n x^n + \cdots + c_1 x + c_0$, the limit laws give

$$\begin{aligned}\lim_{x \rightarrow a} p(x) &= c_n \left(\lim_{x \rightarrow a} x \right)^n + \cdots + c_1 \lim_{x \rightarrow a} x + c_0 \\ &= c_n a^n + \cdots + c_1 a + c_0 = p(a).\end{aligned}$$

If $f(x) = p(x)/q(x)$ is rational and a is in its domain, then $q(a) \neq 0$. By the quotient law,

$$\lim_{x \rightarrow a} f(x) = \frac{\lim_{x \rightarrow a} p(x)}{\lim_{x \rightarrow a} q(x)} = \frac{p(a)}{q(a)} = f(a).$$

□

Three possibilities when we substitute

Consider the same rational function in each example:

$$f(x) = \frac{x^2 - 2x + 1}{x^2 - 1}.$$

Example 1. Find $\lim_{x \rightarrow 2} f(x)$.

Plug in. The denominator is nonzero at 2, so

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x + 1}{x^2 - 1} = \frac{4 - 4 + 1}{4 - 1} = \frac{1}{3}.$$

Example 2. Find $\lim_{x \rightarrow -1} f(x)$.

Substitution gives a numerator tending to 4 and a denominator tending to 0.

Case method. Factor the denominator as $(x - 1)(x + 1)$. The numerator is positive near -1 .

$$x < -1 : (x - 1)(x + 1) > 0 \implies \lim_{x \rightarrow -1^-} f(x) = +\infty,$$

$$x > -1 \text{ near } -1 : (x - 1)(x + 1) < 0 \implies \lim_{x \rightarrow -1^+} f(x) = -\infty.$$

The one-sided limits disagree, so the two-sided limit DNE.

Example 3. Find $\lim_{x \rightarrow 1} f(x)$.

Substitution gives the *indeterminate form* $0/0$: we need to simplify.

Factor and cancel. For $x \neq 1, -1$,

$$\begin{aligned} \frac{x^2 - 2x + 1}{x^2 - 1} &= \frac{(x - 1)^2}{(x - 1)(x + 1)} \\ &= \frac{x - 1}{x + 1}. \end{aligned}$$

Hence

$$\lim_{x \rightarrow 1} \frac{x^2 - 2x + 1}{x^2 - 1} = \lim_{x \rightarrow 1} \frac{x - 1}{x + 1} = \frac{0}{2} = 0.$$

Recap

1. Try plugging in first; check the hypotheses of the limit laws.
2. If the numerator tends to $c \neq 0$ and the denominator tends to 0, check signs on each side. The behavior may be $+\infty$, $-\infty$, or a two-sided limit that does not exist.
3. If substitution gives $0/0$, simplify before trying again.

Practice

For the following three limits, use

$$r(x) = \frac{x^2 - x - 2}{x^2 - 4}.$$

1. Find $\lim_{x \rightarrow 1} r(x)$.

Plug in.

$$\lim_{x \rightarrow 1} \frac{x^2 - x - 2}{x^2 - 4} = \frac{1 - 1 - 2}{1 - 4} = \frac{-2}{-3} = \frac{2}{3}.$$

2. Find $\lim_{x \rightarrow -2} r(x)$.

Case method. The numerator tends to $4 > 0$. Factor the denominator as $(x - 2)(x + 2)$:

$$x \rightarrow -2^- : (x - 2)(x + 2) \rightarrow 0^+ \implies \lim_{x \rightarrow -2^-} r(x) = +\infty,$$

$$x \rightarrow -2^+ : (x - 2)(x + 2) \rightarrow 0^- \implies \lim_{x \rightarrow -2^+} r(x) = -\infty.$$

Therefore the two-sided limit DNE.

3. Find $\lim_{x \rightarrow 2} r(x)$.

Factor and cancel. Substitution gives $0/0$. For $x \neq 2, -2$,

$$\begin{aligned} \frac{x^2 - x - 2}{x^2 - 4} &= \frac{(x - 2)(x + 1)}{(x - 2)(x + 2)} \\ &= \frac{x + 1}{x + 2}. \end{aligned}$$

Thus

$$\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x^2 - 4} = \frac{2 + 1}{2 + 2} = \frac{3}{4}.$$

Indeterminate forms and simplifying

What do we do if substitution gives $0/0$?

Expand. For $h \neq 0$,

$$\begin{aligned}\frac{(2+h)^2 - 4}{h} &= \frac{4 + 4h + h^2 - 4}{h} \\ &= 4 + h.\end{aligned}$$

Therefore $\lim_{h \rightarrow 0} \frac{(2+h)^2 - 4}{h} = 4$.

Remove denominators. For fixed $a \neq 0$,

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h} &= \lim_{h \rightarrow 0} \frac{a - (a+h)}{h a(a+h)} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h a(a+h)} \\ &= \lim_{h \rightarrow 0} \frac{-1}{a(a+h)} = -\frac{1}{a^2}.\end{aligned}$$

The cancellation is valid for $h \neq 0$. Since $a \neq 0$ is fixed, $a+h$ stays nonzero for sufficiently small h .

Conjugate.

$$\begin{aligned}\lim_{x \rightarrow 5} \frac{\sqrt{x+4} - 3}{x-5} &= \lim_{x \rightarrow 5} \frac{(\sqrt{x+4} - 3)(\sqrt{x+4} + 3)}{(x-5)(\sqrt{x+4} + 3)} \\ &= \lim_{x \rightarrow 5} \frac{x+4-9}{(x-5)(\sqrt{x+4} + 3)} \\ &= \lim_{x \rightarrow 5} \frac{1}{\sqrt{x+4} + 3} = \frac{1}{6}.\end{aligned}$$

We cancel $x-5$ only for nearby $x \neq 5$.

Trigonometric identity. For sufficiently small $t \neq 0$,

$$\frac{\sin t}{\tan t} = \frac{\sin t}{\frac{\sin t}{\cos t}} = \cos t.$$

Thus

$$\lim_{t \rightarrow 0} \frac{\sin t}{\tan t} = 1.$$

The small-angle limits used here are justified on page 9.

Practice

Find $\lim_{x \rightarrow 0} \frac{\sqrt{9+x} - 3}{x}$.

Conjugate. For nearby $x \neq 0$,

$$\begin{aligned}\frac{\sqrt{9+x} - 3}{x} &= \frac{9+x-9}{x(\sqrt{9+x}+3)} \\ &= \frac{1}{\sqrt{9+x}+3}.\end{aligned}$$

Therefore

$$\lim_{x \rightarrow 0} \frac{\sqrt{9+x} - 3}{x} = \frac{1}{6}.$$

The squeeze theorem

Theorem (Squeeze theorem). Suppose $\ell(x) \leq f(x) \leq u(x)$ for all sufficiently nearby $x \neq a$. If

$$\lim_{x \rightarrow a} \ell(x) = \lim_{x \rightarrow a} u(x) = L,$$

then $\lim_{x \rightarrow a} f(x) = L$.

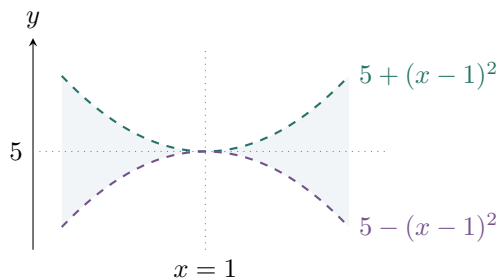
Example 1. Suppose

$$5 - (x-1)^2 \leq f(x) \leq 5 + (x-1)^2$$

near $x = 1$.

Squeeze theorem. The lower and upper bounds both tend to 5 as $x \rightarrow 1$. Hence

$$\lim_{x \rightarrow 1} f(x) = 5.$$



The shaded region contains all permitted values of $f(x)$.

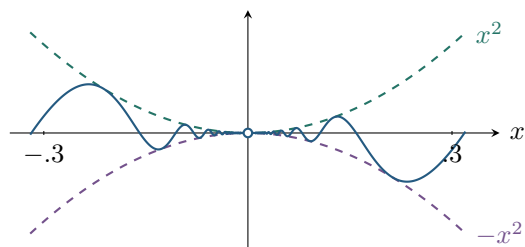
Example 2. Find $\lim_{x \rightarrow 0} x^2 \sin(1/x)$.

Squeeze theorem. Since $-1 \leq \sin(1/x) \leq 1$ and $x^2 \geq 0$,

$$-x^2 \leq x^2 \sin(1/x) \leq x^2 \quad (x \neq 0).$$

Both outer functions tend to 0, so

$$\lim_{x \rightarrow 0} x^2 \sin(1/x) = 0.$$



The oscillations remain between bounds that both approach 0.

Trigonometric limits

Example 3. Find $\lim_{x \rightarrow 0} \frac{\sin(5x)}{2x}$.

Recall $\lim_{u \rightarrow 0} \frac{\sin u}{u} = 1$; its proof follows on page 9. Angles are measured in radians.

Match the argument. Use $u = 5x$, so $u \rightarrow 0$ as $x \rightarrow 0$:

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin(5x)}{2x} &= \frac{5}{2} \lim_{x \rightarrow 0} \frac{\sin(5x)}{5x} \\ &= \frac{5}{2} \lim_{u \rightarrow 0} \frac{\sin u}{u} \\ &= \frac{5}{2}.\end{aligned}$$

Difficult

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x}$$

Conjugate. Multiply by $(1 + \cos x)/(1 + \cos x)$:

$$\begin{aligned}\frac{1 - \cos x}{x} &= \frac{(1 - \cos x)(1 + \cos x)}{x(1 + \cos x)} \\ &= \frac{1 - \cos^2 x}{x(1 + \cos x)} \\ &= \frac{\sin^2 x}{x(1 + \cos x)} \\ &= \frac{\sin x}{x} \cdot \frac{\sin x}{1 + \cos x}.\end{aligned}$$

Using the trigonometric limits established in the following proof,

$$\frac{\sin x}{x} \rightarrow 1, \quad \sin x \rightarrow 0, \quad 1 + \cos x \rightarrow 2.$$

Therefore

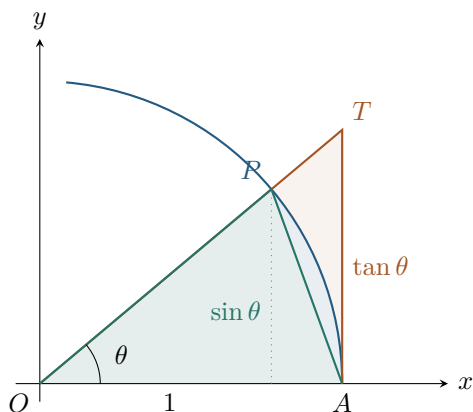
$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 1 \cdot \frac{0}{2} = 0.$$

The fundamental trigonometric limit

Theorem. For angles measured in radians,

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.$$

Proof. First suppose $0 < \theta < \pi/2$. On the unit circle, let $O = (0, 0)$, $A = (1, 0)$, and $P = (\cos \theta, \sin \theta)$. The ray OP meets the line $x = 1$ at $T = (1, \tan \theta)$.



Comparing the areas of triangle OAP , sector OAP , and triangle OAT gives

$$\frac{1}{2} \sin \theta \leq \frac{1}{2} \theta \leq \frac{1}{2} \tan \theta, \quad \text{so} \quad \sin \theta \leq \theta \leq \tan \theta.$$

Since $0 \leq \sin \theta \leq \theta$, we have $\sin \theta \rightarrow 0$ as $\theta \rightarrow 0^+$. Odd symmetry gives the left limit. Near 0, cosine is positive, so

$$\lim_{\theta \rightarrow 0} \cos \theta = \lim_{\theta \rightarrow 0} \sqrt{1 - \sin^2 \theta} = 1.$$

Now take reciprocals of the positive quantities in $\sin \theta \leq \theta \leq \tan \theta$, then multiply by $\sin \theta$:

$$\begin{aligned} \frac{\cos \theta}{\sin \theta} &\leq \frac{1}{\theta} \leq \frac{1}{\sin \theta} \\ \implies \cos \theta &\leq \frac{\sin \theta}{\theta} \leq 1. \end{aligned}$$

Both bounds approach 1, so the squeeze theorem gives

$$\lim_{\theta \rightarrow 0^+} \frac{\sin \theta}{\theta} = 1.$$

For the left-hand limit, put $u = -\theta$. Then $u \rightarrow 0^+$ and

$$\lim_{\theta \rightarrow 0^-} \frac{\sin \theta}{\theta} = \lim_{u \rightarrow 0^+} \frac{\sin(-u)}{-u} = \lim_{u \rightarrow 0^+} \frac{\sin u}{u} = 1.$$

The two one-sided limits agree, proving the theorem. □